Bijay Thakur

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SUMMARY

AI Engineer specializing in production RAG and multi-agent orchestration. Shipped a live financial QA platform with Supabase auth, Stripe-gated quotas, and Google Cloud Run end-to-end across OpenAI, Gemini, LangGraph, LlamaIndex, ChromaDB, and FastAPI.

EDUCATION

Queens College, The City University of New York **Expected: Dec 2026**
Bachelor of Science in Computer Science, ***GPA:*** *3.64*

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, C++, SQL
- **AI & ML:** RAG Pipelines, Hybrid Retrieval (BM25 + RRF), Cross-Encoder Reranking, Multi-Agent Orchestration, LLM Fine-Tuning, LangGraph, LlamaIndex, Hugging Face Transformers, PyTorch
- **Backend/DevOps:** FastAPI, Flask, Express, REST APIs, Docker, Google Cloud Run, GitHub Actions

PROJECTS

SEC Insight AI: Production Financial QA System | Live Demo **April 2026 - June 2026**

- Engineered a citation-grounded RAG pipeline over Apple's 2025 annual report with OpenAI embeddings, ChromaDB, BM25 hybrid retrieval, RRF, and cross-encoder reranking — achieving 0.9901 RAGAS faithfulness and 0.8032 MRR on a 50-question gold eval set.
- Deployed full-stack to production: Dockerized FastAPI backend on Google Cloud Run, Next.js frontend on Vercel, Supabase auth with per-user quota enforcement, tiered rate limiting (30–300 req/hr), answerability gate, and server-side cost guards — with 9-file test suite and CI/CD via GitHub Actions.

RivalScope AI: Agentic Competitive Intelligence Platform | GitHub **Jun 2026 - July 2026**

- Engineered an agentic competitive-intelligence platform using Next.js 15, LangGraph, LangChain, Tavily, LangSmith tracing, and SSE streaming to run live competitor research and generate cited GTM reports.
- Built an evaluation-driven grounding layer with Pydantic structured outputs, citation checks, LLM-as-judge evals, and cost/context guards — achieving 88.9% comparison grounding on the latest real-mode benchmark.

LaunchLab AI: Agentic Multi-LLM Product Studio | GitHub **April 2026 - May 2026**

- Architected a multi-LLM agentic system using LangGraph, Flask, Gemini, and OpenAI to orchestrate 7 specialized agents — producing market research, product blueprints, Claude Code build prompts, campaign copy, visual prompts, and evaluation reports from a single product brief.
- Implemented manual v1 and LangGraph v2 orchestration modes with identical response shapes, provider fallback logic (live → partial → mock), and structured error handling - enabling direct latency and reliability benchmarking between orchestration approaches.

Code Council: Multi-Agent GenAI Code Analysis | GitHub **Jan 2026 - May 2026**

- Built a 6-agent code analysis system (Orchestrator, Explainer, Bug Hunter, Complexity Analyst, Debate, Supervisor) using Express and React/TypeScript, with Explainer, Bug Hunter, and Complexity agents executing in parallel via Promise.all
- Implemented Zod schema validation on all LLM outputs, trace ID logging, fallback-safe responses, and a context-aware chat assistant — preventing malformed LLM responses from breaking downstream agent state.

MarianMT Domain Adaptation: Neural Machine Translation Fine-Tuning **May 2026**

- Fine-tuned Helsinki-NLP/opus-mt-en-fr using Hugging Face Seq2SeqTrainer, PyTorch, and mixed-precision CUDA on Google Colab across 1K, 10K, and 189K KDE4 training examples — improving English-to-French BLEU from 46.07 to 58.37 (+26.7%) and validating consistent scaling improvement across all dataset sizes.

EXPERIENCE

Academic Tutor, LaGuardia Community College **Feb 2024 - Jun 2024**
Long Island City, NY

- Tutored students 1-on-1 in C++, debugging, and algorithmic problem-solving; developed structured practice modules to accelerate mastery of core CS fundamentals.